TASK 5

PART A - CYBERCRIME AWARENESS CASE STUDY

Case Study: Ransomware Attack on Colonial Pipeline

Ransomware is a type of malicious software (malware) designed to block access to a computer system or encrypt its data until a sum of money is paid. In modern "double extortion" schemes, attackers also steal sensitive data and threaten to leak it publicly if the ransom is ignored.

In May 2021, the Colonial Pipeline, which provides 45% of the U.S. East Coast's fuel, fell victim to the DarkSide hacking group. Hackers gained entry using a single compromised password for an old Virtual Private Network (VPN) account that lacked multi-factor authentication (MFA), and they then stole a lot of data. The ransomware payload was executed, encrypting the company's billing and accounting systems. In exchange for a decryption tool, they asked for millions worth of Bitcoin.

Ransomware typically targets critical infrastructure, healthcare, and large enterprises because these organizations cannot afford downtime.

The consequences for Colonial Pipeline were massive because the company faced a huge financial loss of millions, and operations had to be shut down as well, leading to fuel shortages and a state of emergency.

The key lesson is that most ransomware attacks are actually quite simple because they just exploit the easy passwords people set once and never change. And people also tend to reuse passwords, and this worsens the situation. Using multi-factor authentication is the only way we can prevent this.

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